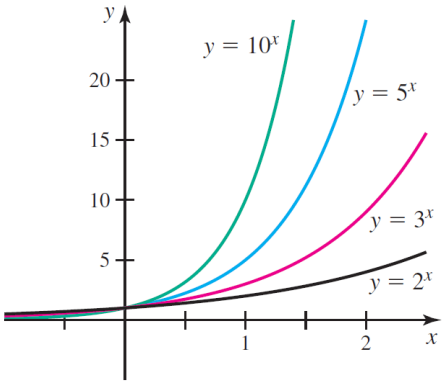
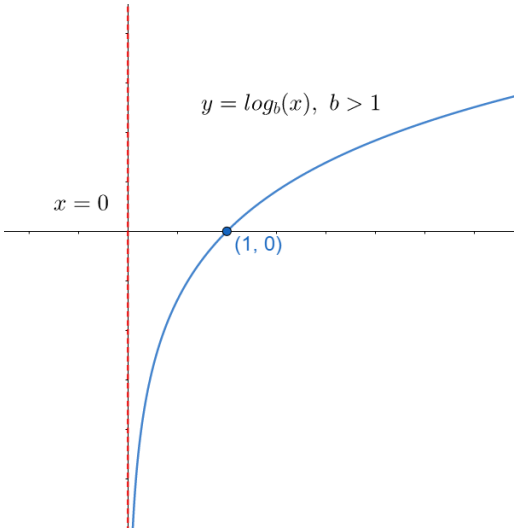
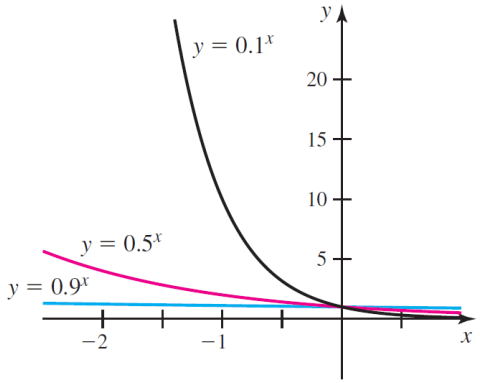
$y = \log_b x$, $b > 1$ $Domain = (0, \infty)$ $\quad$ $Range = (-\infty, \infty)$ Increasing on $(0, \infty)$ x-intercept: $(1, 0)$ $x \to \infty \Rightarrow y = \log_b x \to \infty$ $x \to 0^+ \Rightarrow y = \log_b x \to -\infty \Rightarrow x = 0$ is a vertical asymptote **Note:** $y = \ln x = \log_e x$ where $e \approx 2.7182818$	$y = b^x$, $b > 1$ $Domain = (-\infty, \infty)$ $\quad$ $Range = (0, \infty)$ Increasing on $(-\infty, \infty)$ y-intercept: $(0, 1)$ $x \to \infty \Rightarrow y = b^x \to \infty$ $x \to -\infty \Rightarrow y = b^x \to 0 \Rightarrow y = 0$ is a horizontal asymptote
$y = \log_b x$, $0 < b < 1$ $Domain = (0, \infty)$ $\quad$ $Range = (-\infty, \infty)$ Decreasing on $(0, \infty)$ x-intercept: $(1, 0)$ $x \to \infty \Rightarrow y = \log_b x \to -\infty$ $x \to 0^+ \Rightarrow y = \log_b x \to \infty \Rightarrow x = 0$ is a vertical asymptote	$y = b^x$, $0 < b < 1$ $Domain = (-\infty, \infty)$ $\quad$ $Range = (0, \infty)$ Decreasing on $(-\infty, \infty)$ y-intercept: $(0, 1)$ $x \to \infty \Rightarrow y = b^x \to 0 \Rightarrow y = 0$ is a horizontal asymptote $x \to -\infty \Rightarrow y = b^x \to \infty$

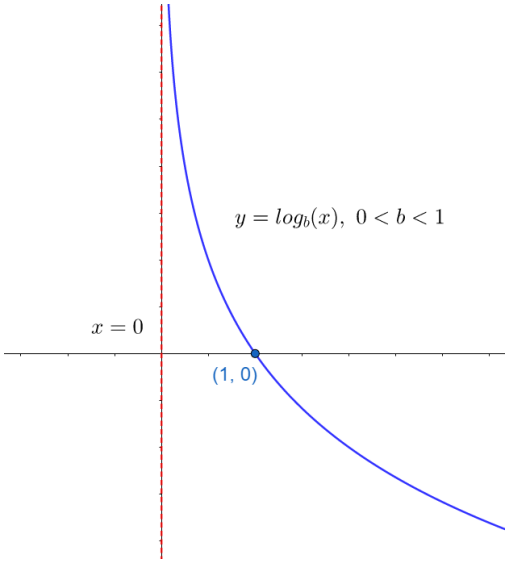
Larger values of b produce greater rates of increase in b^x if $b > 1$.





Smaller values of b produce greater rates of decrease in b^x if $0 < b < 1$.





The **Floor Function** $f(x) = \lfloor x \rfloor$ is a function which gives the greatest integer less than or equal to x.

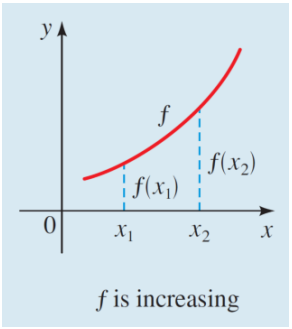
$Domain = (-\infty, \infty)$

$Range = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\} = \mathbb{Z}$

Increasing on $(-\infty, \infty)$

$x \to \infty \Rightarrow f(x) = \lfloor x \rfloor \to \infty$ $\quad$ $x \to -\infty \Rightarrow f(x) = \lfloor x \rfloor \to -\infty$

Definition: A function is called increasing on an interval if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for any x_1 and x_2 in the interval.



f is increasing

Can you think of a function that is increasing on its domain?

The **Ceiling Function** $f(x) = \lceil x \rceil$ is a function which gives the smallest integer greater than or equal to x.

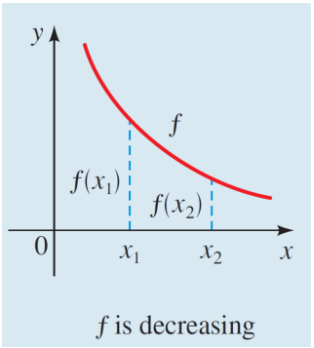
$Domain = (-\infty, \infty)$

$Range = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\} = \mathbb{Z}$

Increasing on $(-\infty, \infty)$

$x \to \infty \Rightarrow f(x) = \lceil x \rceil \to \infty$ $\quad$ $x \to -\infty \Rightarrow f(x) = \lceil x \rceil \to -\infty$

Definition: A function is called decreasing on an interval if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$ for any x_1 and x_2 in the interval.



f is decreasing

Can you think of a function that is decreasing on its domain?

The following functions are increasing on their domains:

$y = \log_b x \ , \ b > 1$

$y = b^x \ , \ b > 1$

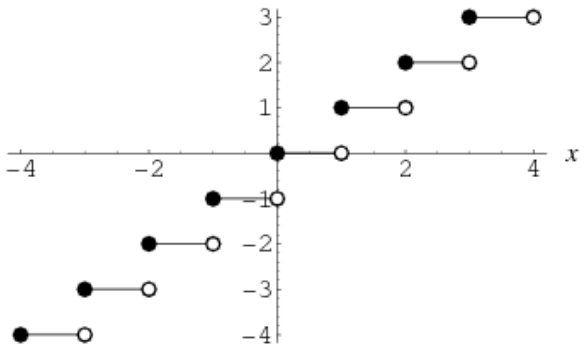
$y = x^n$, where n is odd: $y = x \ , y = x^3 , \ldots$

$y = \sqrt[n]{x} \ : \ y = \sqrt{x} \ , y = \sqrt[3]{x} \ , \ldots$

$y = \sin^{-1} x$

$y = \tan^{-1} x$

$f(x) = \lfloor x \rfloor$



The following functions are decreasing on their domains:

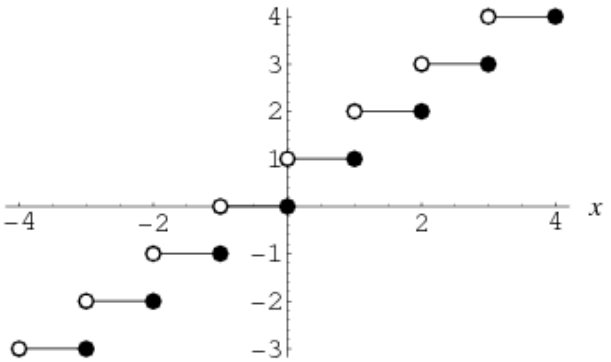
$y = \log_b x \ , \ 0 < b < 1$

$y = b^x \ , \ 0 < b < 1$

$y = \cos^{-1} x$

$y = \cot^{-1} x$

$f(x) = \lceil x \rceil$



The **absolute value function**: $f(x)=|x|$

$Domain=(-\infty,\infty)$ $Range=[0,\infty)$

Increasing on $(0,\infty)$

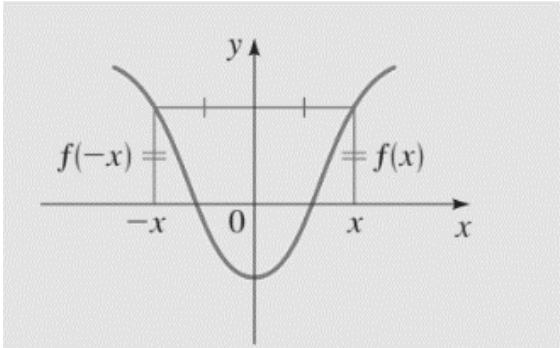
Decreasing on $(-\infty,0)$

x and y-intercept: $(0,0)$

$x\to\infty \Rightarrow f(x)=|x|\to\infty$

$x\to-\infty \Rightarrow f(x)=|x|\to\infty$

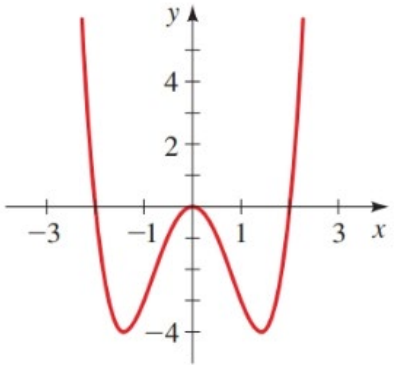
f is <u>even</u> if $f(-x)=f(x)$ for all x in the domain of f.



The graph of an even function is symmetric with respect to the y-axis.

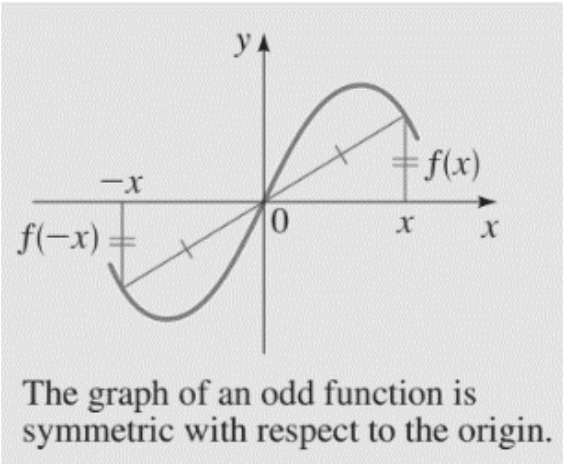
Can you think of examples of even functions?

You are given the graph of a function $y=f(x)$.



What would the graph of $g(x)=|f(x)|$ look like?

f is <u>odd</u> if $f(-x)=-f(x)$ for all x in the domain of f.



The graph of an odd function is symmetric with respect to the origin.

Can you think of examples of odd functions?

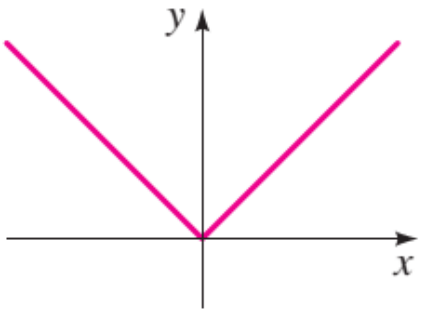
The following functions are even:

$y = x^2$

$y = \dfrac{1}{x^2}$

$y = \cos x$

$y = \sec x$



$f(x) = |x|$

The following functions are odd:

$y = x^3$

$y = \dfrac{1}{x}$

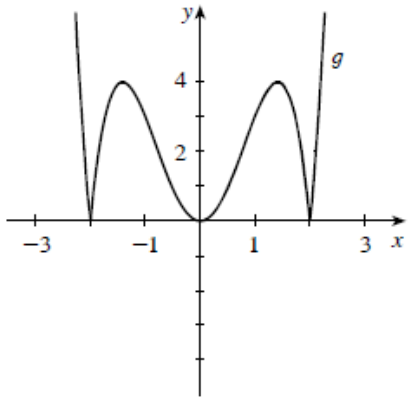
$y = \sin x$

$y = \csc x$,

$y = \tan x$

$y = \cot x$

Graph of $g(x) = |x|$.



Rational Function: $y = \frac{1}{x}$

$Domain = (-\infty, 0) \cup (0, \infty)$ $Range = (-\infty, 0) \cup (0, \infty)$

Decreasing on $(-\infty, 0) \cup (0, \infty)$

$x \to \infty \Rightarrow y = \frac{1}{x} \to 0$, $x \to -\infty \Rightarrow y = \frac{1}{x} \to 0$

$\Rightarrow y = 0$ is a horizontal asymptote

$x \to 0^+ \Rightarrow y = \frac{1}{x} \to \infty$, $x \to 0^- \Rightarrow y = \frac{1}{x} \to -\infty$

$\Rightarrow x = 0$ is a vertical asymptote

Power Function: $y = x^n$, n is even

$Domain = (-\infty, \infty)$ $Range = [0, \infty)$

Increasing on $(0, \infty)$ Decreasing on $(-\infty, 0)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = x^n \to \infty$ $x \to -\infty \Rightarrow y = x^n \to \infty$

Even Function.

Symmetric with respect to the *y*-axis.

Rational Function: $y = \frac{1}{x^2}$

$Domain = (-\infty, 0) \cup (0, \infty)$ $Range = (0, \infty)$

Increasing on $(-\infty, 0)$ Decreasing on $(0, \infty)$

$x \to \infty \Rightarrow y = \frac{1}{x^2} \to 0$, $x \to -\infty \Rightarrow y = \frac{1}{x^2} \to 0$

$\Rightarrow y = 0$ is a horizontal asymptote

$x \to 0^+ \Rightarrow y = \frac{1}{x^2} \to \infty$, $x \to 0^- \Rightarrow y = \frac{1}{x^2} \to \infty$

$\Rightarrow x = 0$ is a vertical asymptote

Power Function: $y = x^n$, n is odd

$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

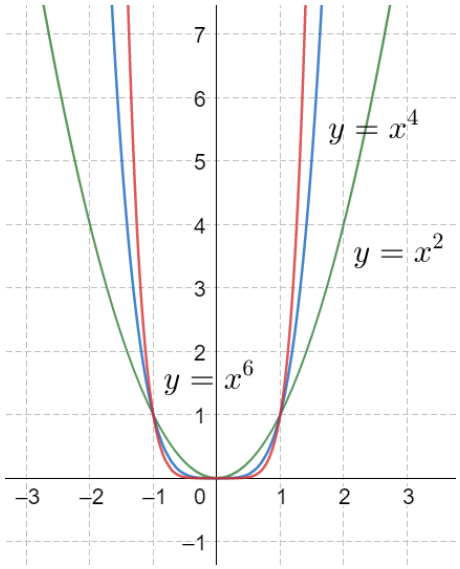
Increasing on $(-\infty, \infty)$

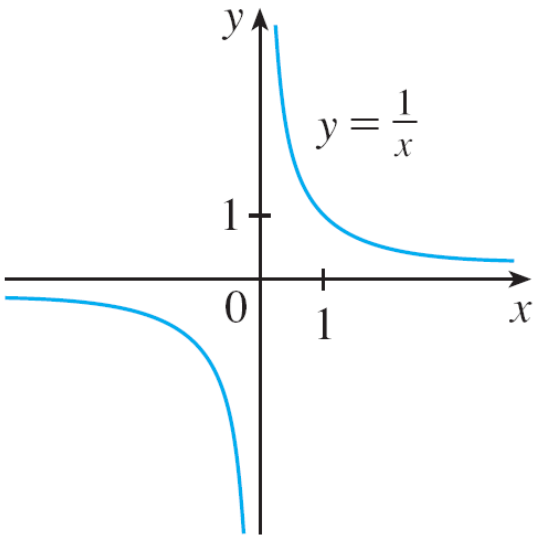
x and y-intercept: $(0,0)$

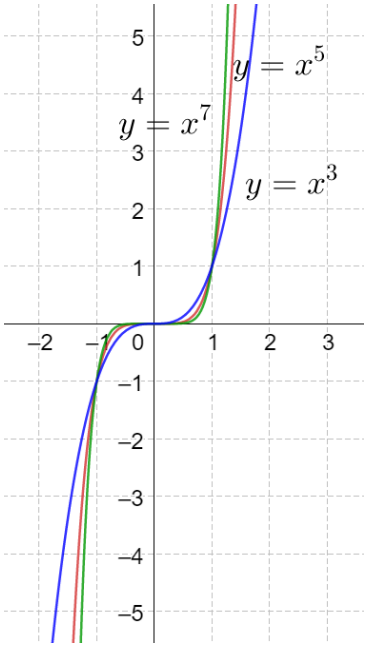
$x \to \infty \Rightarrow y = x^n \to \infty$ $x \to -\infty \Rightarrow y = x^n \to -\infty$

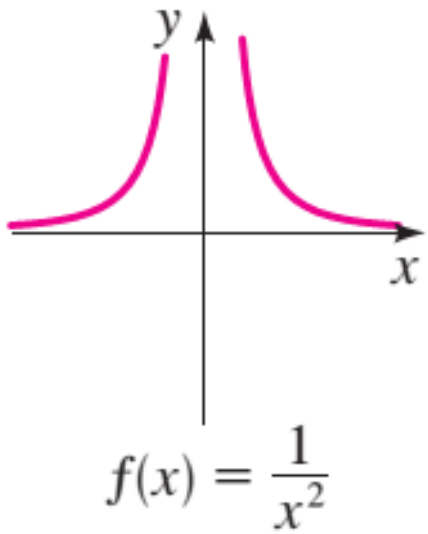
Odd function.

Symmetric with respect to the origin.


$y = x^4$
$y = x^2$
$y = x^6$


y
$y = \frac{1}{x}$
1
0
1
x


$y = x^5$
$y = x^7$
$y = x^3$


y
x
$f(x) = \frac{1}{x^2}$

Radical Function: $y = \sqrt[n]{x} = x^{1/n}$, n is even

$Domain = [0, \infty)$ $Range = [0, \infty)$

Increasing on $(0, \infty)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = \sqrt[n]{x} \to \infty$

Linear Function: $y = mx + b$, $m > 0$

$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

If $m > 0$, then $y = mx + b$ is increasing on $(-\infty, \infty)$

x-intercept: $\left(-\frac{b}{m}, 0\right)$ y-intercept: $(0, b)$

$x \to \infty \Rightarrow y = mx + b \to \infty$

$x \to -\infty \Rightarrow y = mx + b \to -\infty$

Radical Function: $y = \sqrt[n]{x} = x^{1/n}$, n is odd

$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

Increasing on $(-\infty, \infty)$

x and y-intercept: $(0,0)$

$x \to \infty \Rightarrow y = \sqrt[n]{x} \to \infty$ $x \to -\infty \Rightarrow y = \sqrt[n]{x} \to -\infty$

Odd function.

Symmetric with respect to the origin.

Linear Function: $y = mx + b$, $m < 0$

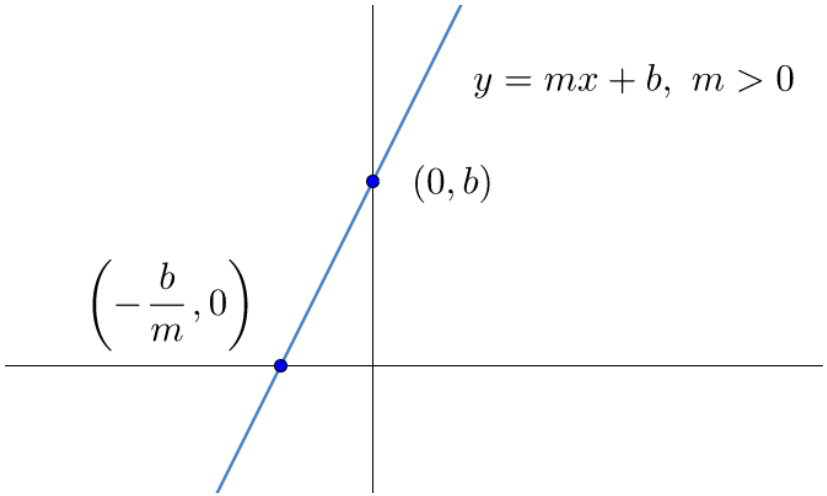
$Domain = (-\infty, \infty)$ $Range = (-\infty, \infty)$

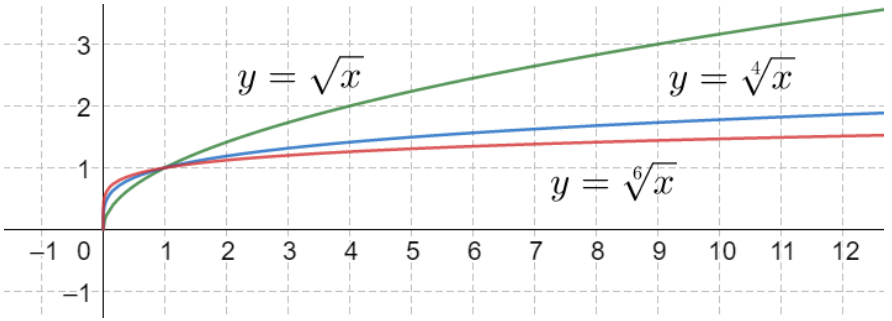
If $m < 0$, then $y = mx + b$ is decreasing on $(-\infty, \infty)$

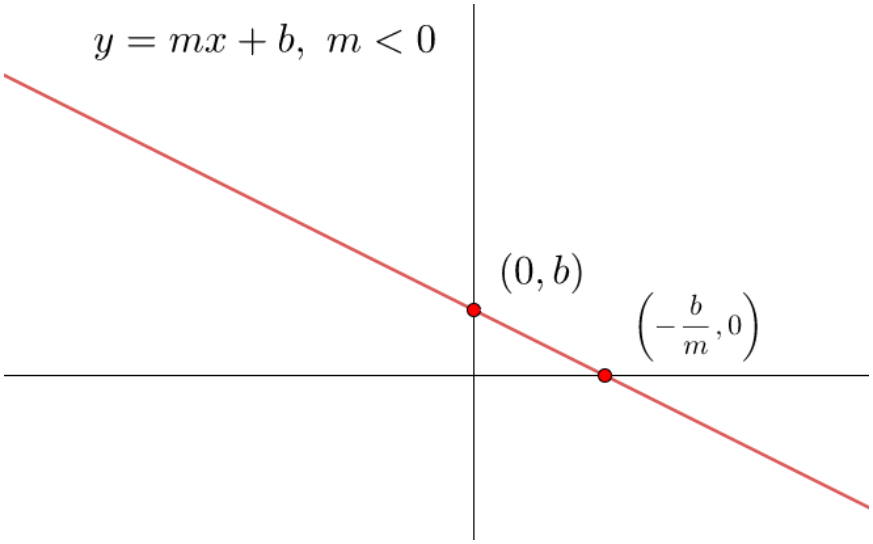
x-intercept: $\left(-\frac{b}{m}, 0\right)$ y-intercept: $(0, b)$

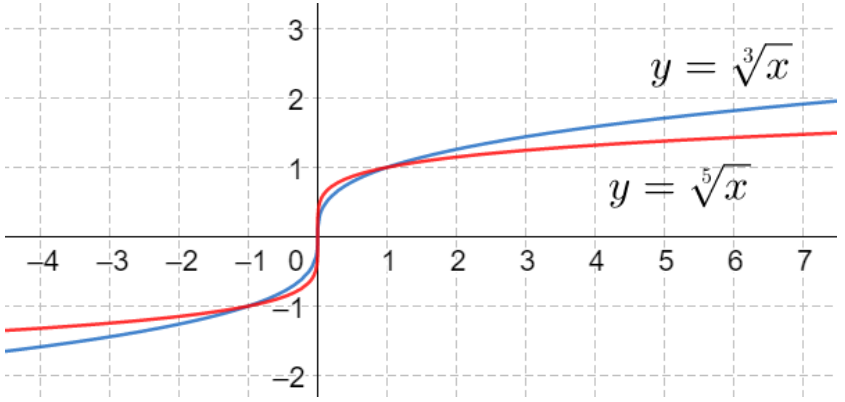
$x \to \infty \Rightarrow y = mx + b \to -\infty$

$x \to -\infty \Rightarrow y = mx + b \to \infty$


$y = mx + b,\ m > 0$
$(0, b)$
$\left(-\frac{b}{m}, 0\right)$


$y = \sqrt{x}$
$y = \sqrt[4]{x}$
$y = \sqrt[6]{x}$


$y = mx + b,\ m < 0$
$(0, b)$
$\left(-\frac{b}{m}, 0\right)$


$y = \sqrt[3]{x}$
$y = \sqrt[5]{x}$

$y = b$

$Domain = (-\infty, \infty)$ $Range = \{b\}$

$y = b$ is constant on $(-\infty, \infty)$

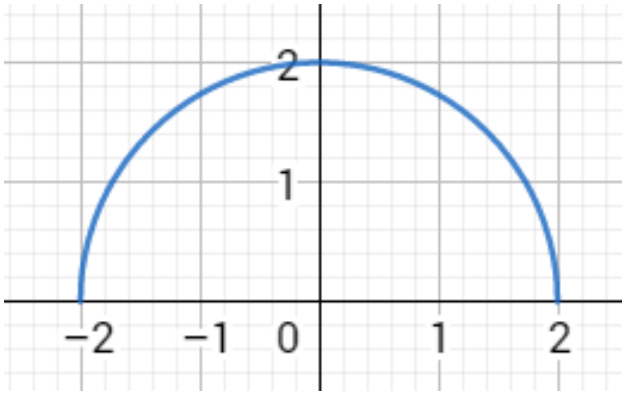
y-intercept: $(0, b)$

Semi-Circle (Upper) centered at Origin: $y = \sqrt{r^2 - x^2}$

Example: $y = \sqrt{4 - x^2}$

Domain: $[-2, 2]$

Range $[0, 2]$

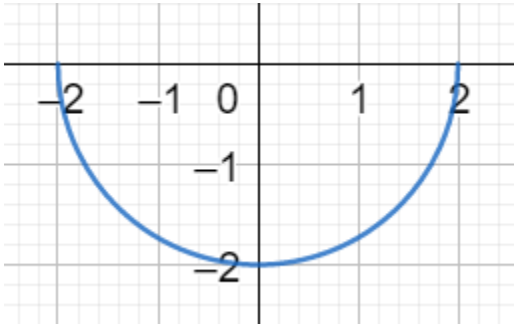


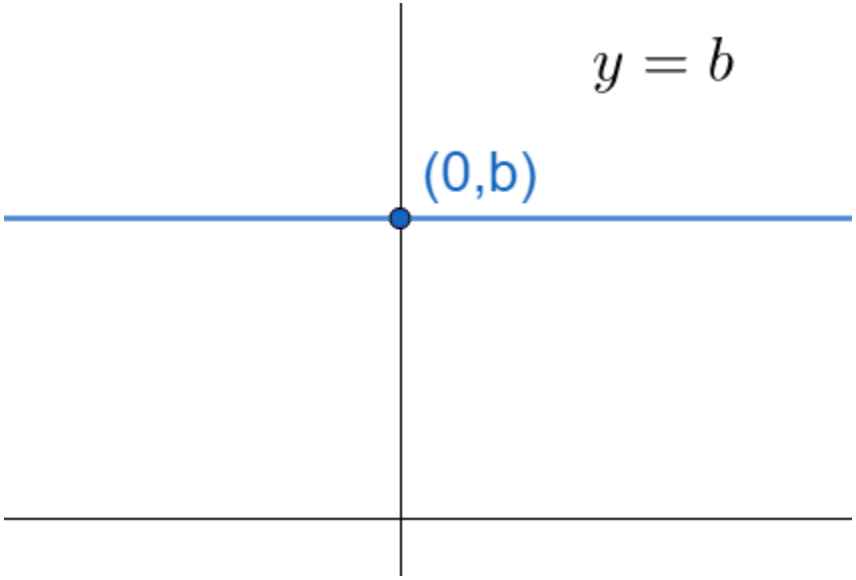
Semi-Circle(Lower) centered at Origin: $y = -\sqrt{r^2 - x^2}$

Example: $y = -\sqrt{4 - x^2}$

Domain: $[-2, 2]$

Range $[-2, 0]$





$y = \sin(x)$

$Domain = (-\infty, \infty)$ $Range = [-1,1]$

$Period = 2\pi$

x-intercepts: $(\pi k, 0)$ where k is an integer

y-intercept: $(0,0)$

Odd function.

Symmetric with respect to the origin.

$y = \csc(x)$

$Domain = \{x \in \mathbb{R} | x \neq \pi k \text{ where } k \text{ is an integer}\}$

$Range = (-\infty, -1] \cup [1, \infty)$

$Period = 2\pi$

vertical asymptotes: $x = \pi k$ where k is an integer

Odd function.

Symmetric with respect to the origin.

$y = \cos(x)$

$Domain = (-\infty, \infty)$ $Range = [-1,1]$

$Period = 2\pi$

x-intercepts: $\left(\frac{\pi}{2}(2k+1), 0\right)$ where k is an integer

y-intercept: $(0,1)$

Even function.

Symmetric with respect to the y-axis.

$y = \sec(x)$

$Domain = \left\{x \in \mathbb{R} | x \neq \frac{\pi}{2}(2k+1) \text{ where } k \text{ is an integer}\right\}$

$Range = (-\infty, -1] \cup [1, \infty)$

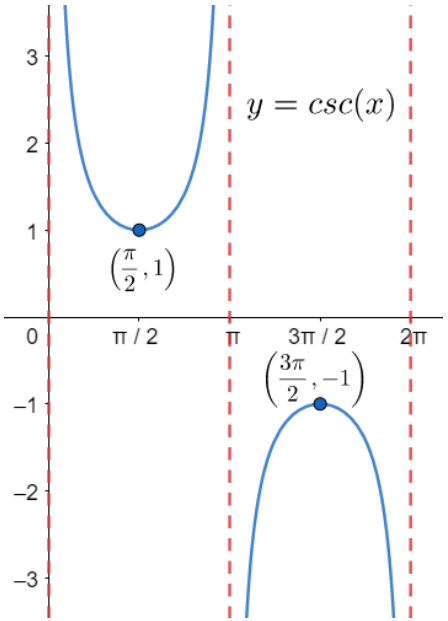
$Period = 2\pi$

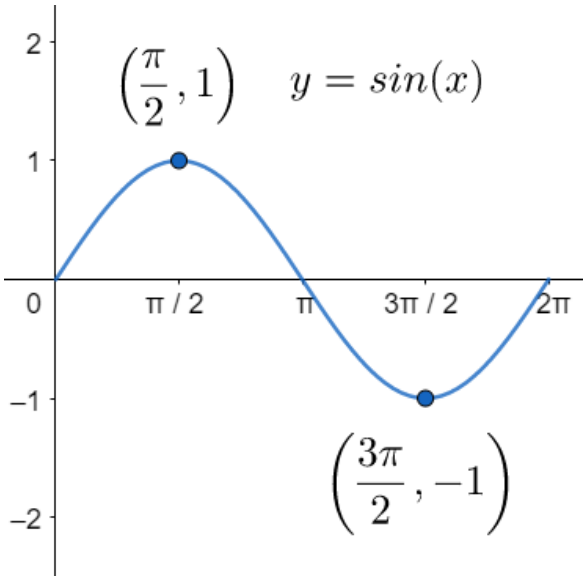
y-intercept: $(0,1)$

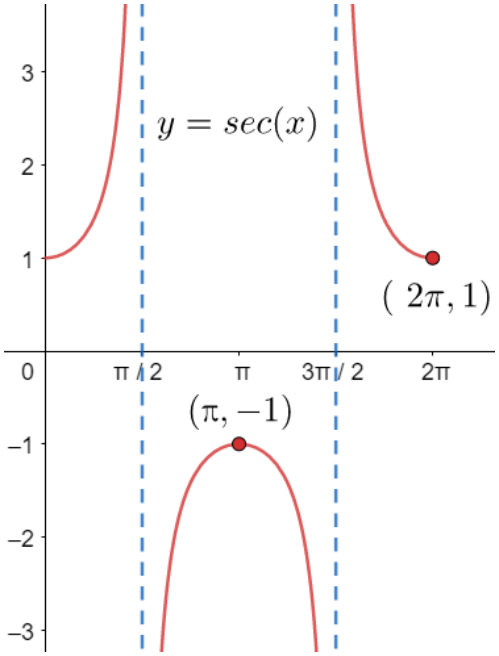
vertical asymptotes: $x = \frac{\pi}{2}(2k+1)$ where k is an integer

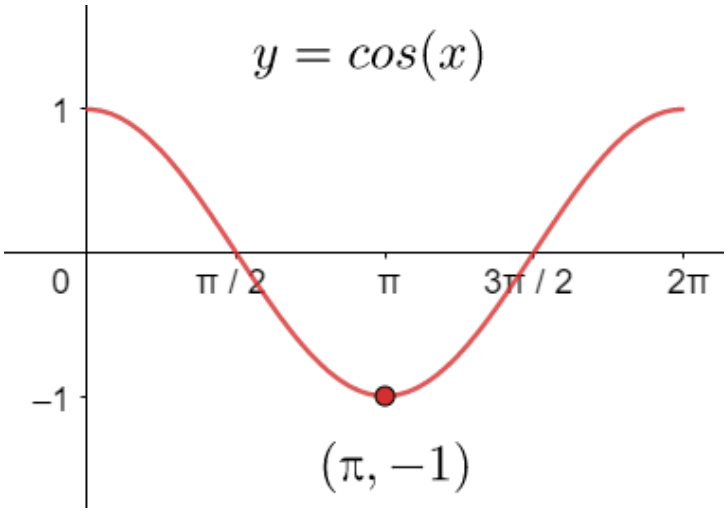
Even function.

Symmetric with respect to the y-axis.


$y = csc(x)$
$\left(\frac{\pi}{2}, 1\right)$
$\left(\frac{3\pi}{2}, -1\right)$
3
2
1
0
−1
−2
−3
π / 2
π
3π / 2
2π


$\left(\frac{\pi}{2}, 1\right)$
$y = sin(x)$
$\left(\frac{3\pi}{2}, -1\right)$
2
1
0
−1
−2
π / 2
π
3π / 2
2π


$y = sec(x)$
$(2\pi, 1)$
$(\pi, -1)$
3
2
1
0
−1
−2
−3
π / 2
π
3π / 2
2π


$y = cos(x)$
$(\pi, -1)$
1
0
−1
π / 2
π
3π / 2
2π

$y = \tan(x)$

$Domain = \left\{ x \in \mathbb{R} \middle| x \neq \frac{\pi}{2}(2k+1) \text{ where } k \text{ is an integer} \right\}$

$Range = (-\infty, \infty)$

$Period = \pi$

vertical asymptotes: $x = \frac{\pi}{2}(2k+1)$ where k is an integer

x-intercepts: $(\pi k, 0)$ where k is an integer

y-intercept: $(0,0)$

Odd function. Symmetric with respect to the origin.

$y = \sin^{-1} x$

$Domain = [-1,1]$ $Range = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$

x-intercepts: $(0,0)$

y-intercept: $(0,0)$

$y = \cot(x)$

$Domain = \{ x \in \mathbb{R} | x \neq \pi k \text{ where } k \text{ is an integer} \}$

$Range = (-\infty, \infty)$

$Period = \pi$

vertical asymptotes: $x = \pi k$ where k is an integer

x-intercepts: $\left(\frac{\pi}{2}(2k+1), 0 \right)$ where k is an integer

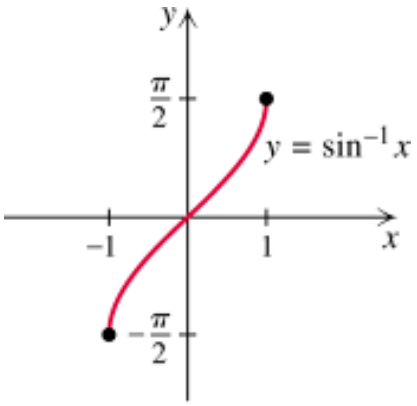
Odd function. Symmetric with respect to the origin.

$y = \cos^{-1} x$

$Domain = [-1,1]$ $Range = [0, \pi]$

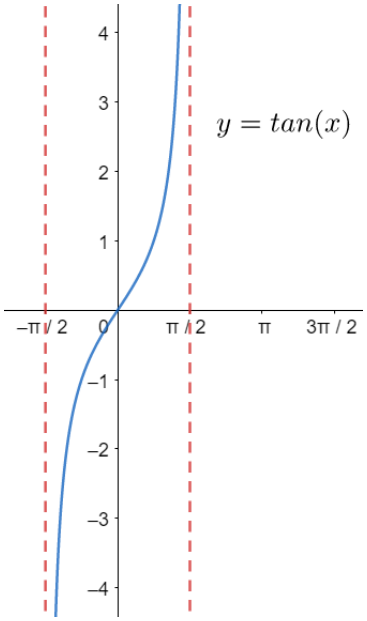
x-intercepts: $(1,0)$

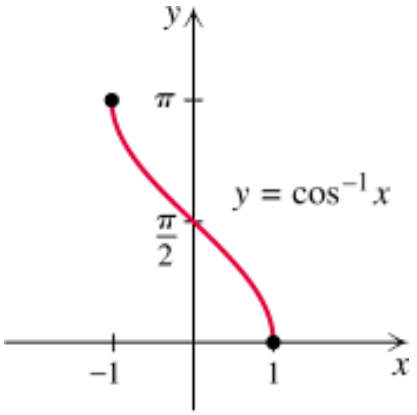
y-intercept: $\left(0, \frac{\pi}{2} \right)$



Domain [−1, 1]

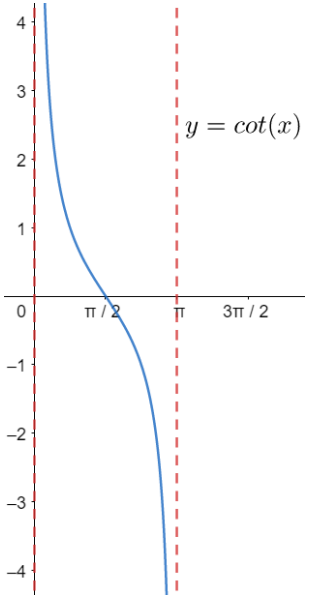
Range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$





Domain [−1, 1]

Range [0, π]



$y = \csc^{-1} x$

$Domain = (-\infty, -1] \cup [1, \infty)$ $\quad$ $Range = \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

$x \to \infty \Rightarrow y = \csc^{-1} x \to 0$

$x \to -\infty \Rightarrow y = \csc^{-1} x \to 0$

$y = 0$ is a horizontal asymptote

$y = \tan^{-1} x$

$Domain = (-\infty, \infty)$

$Range = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

x-and y-intercept: $(0, 0)$

$x \to \infty \Rightarrow y = \tan^{-1} x \to \frac{\pi}{2} \quad \Rightarrow \quad y = \frac{\pi}{2}$ is a horizontal asymptote

$x \to -\infty \Rightarrow y = \tan^{-1} x \to -\frac{\pi}{2} \quad \Rightarrow \quad y = -\frac{\pi}{2}$ is a horizontal asymptote

$y = \sec^{-1} x$

$Domain = (-\infty, -1] \cup [1, \infty)$ $\quad$ $Range = \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

x-intercepts: $(1, 0)$

$x \to \infty \Rightarrow y = \sec^{-1} x \to \frac{\pi}{2}$

$x \to -\infty \Rightarrow y = \sec^{-1} x \to \frac{\pi}{2}$

$y = \frac{\pi}{2}$ is a horizontal asymptote

$y = \cot^{-1} x$

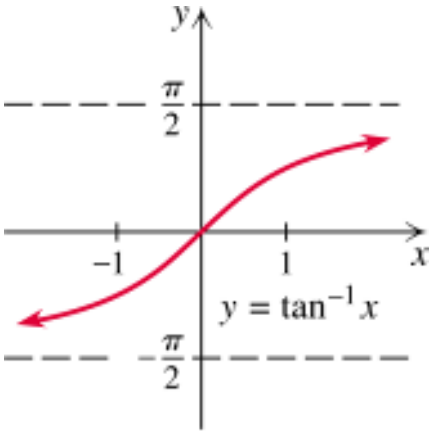
$Domain = (-\infty, \infty)$

$Range = (0, \pi)$

y-intercept: $\left(0, \frac{\pi}{2}\right)$

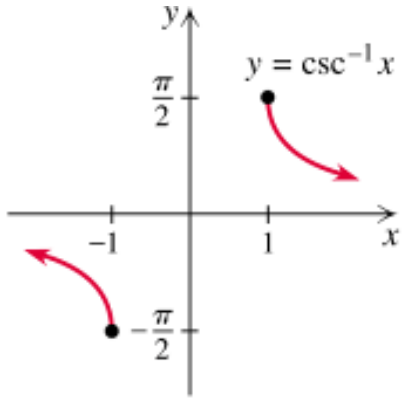
$x \to \infty \Rightarrow y = \cot^{-1} x \to 0 \quad \Rightarrow \quad y = 0$ is a horizontal asymptote

$x \to -\infty \Rightarrow y = \cot^{-1} x \to \pi \quad \Rightarrow \quad y = \pi$ is a horizontal asymptote



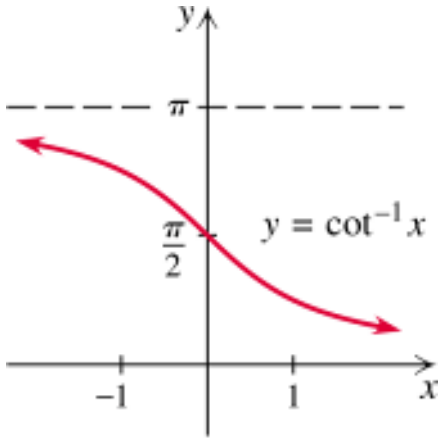
Domain $(-\infty, \infty)$

Range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



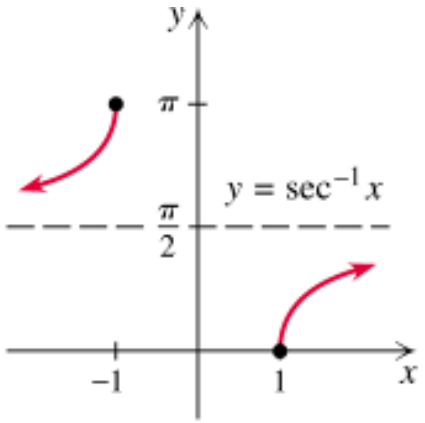
Domain $(-\infty, -1] \cup [1, \infty)$

Range $\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$



Domain $(-\infty, \infty)$

Range $(0, \pi)$



Domain $(-\infty, -1] \cup [1, \infty)$

Range $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$